

# TrES-5b

R-0947

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_TRES-5\_230621 · created 2026-07-29T13:50:33+00:00

## BENCHMARK: MARGINAL

### Results

|                                                   |                                 |
|---------------------------------------------------|---------------------------------|
| Mid-Transit Time (Tmid)                           | 2460116.7829 +/- 0.0031 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.184 +/- 0.043                 |
| Transit depth (Rp/Rs)^2                           | 3.37 +/- 1.57 [%]               |
| Orbital Inclination (inc)                         | 82.06 +/- 2.47                  |
| Airmass coefficient 1 (a1)                        | 1.039 +/- 0.015                 |
| Airmass coefficient 2 (a2)                        | -0.023 +/- 0.038                |
| Scatter in the residuals of the lightcurve fit is | 1.48 %                          |
| Best Comparison Star                              | #2 - [280, 366]                 |
| Optimal Method                                    | PSF photometry                  |
| Transit Duration (day)                            | 0.037 +/- 0.016                 |

### Accuracy vs published ephemeris (ExoClock/NEA)

|                                      |                                                       |
|--------------------------------------|-------------------------------------------------------|
| Rp/R $\blacksquare$ fit vs published | 0.184 $\pm$ 0.043 vs 0.142 (deviation 0.98 $\sigma$ ) |
| Duration fit vs published            | 0.037 d vs 0.0758 d (deviation 2.43 $\sigma$ )        |
| Mid-time O–C                         | 8.8 min                                               |
| Depth SNR                            | 4.3                                                   |
| Residual scatter                     | 1.48 %                                                |

### Light curve

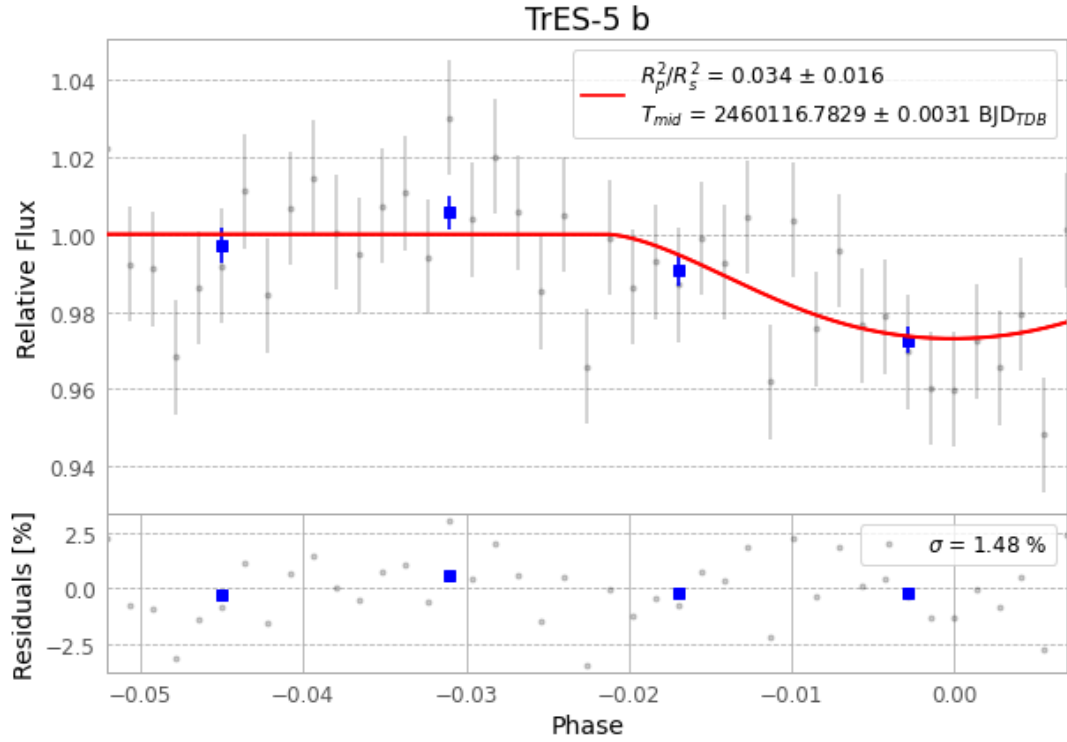


Figure 1 — FinalLightCurve\_TrES-5 b\_2023-06-20.png (SHA-256 393b09a45f1f16a7...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [297, 228], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 25d1fcd3941062d8...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit ead60b9

## Data provenance

43 FITS frames (28 MB), TRES-5230621045415.FITS → TRES-5230621070015.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-230621.log (22005b7db4f1...), FinalLightCurve\_TrES-5 b\_2023-06-20.png (393b09a45f1f...), FinalLightCurve\_TrES-5 b\_2023-06-20.csv (e8a21686f532...)

## Compute resources

Wall time 120.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

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Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 13:50Z.